

Solar Sample Portfolio (3 sites)

Solar PV Farm / Rooftop Array · 2026-09-30

OVERALL CONDITION POOR	7 Findings	2 P1 Immediate	3 P2 Planned	2 P3 Monitor
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Source footage	solar_demo_set
Footage length	Still images
Frames analysed	10
Raw observations	8 (0 rejected as likely false alarms)
Detection model	Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)
Triage model	TypeSafe Jev (jev-1.13.0)
Processing time	2.1 s

How urgent is each priority

P1	Immediate (within 24-72 h)
P2	Planned (within 30 days)
P3	Scheduled / monitor (next maintenance cycle)

1. Executive summary

This sample covers three solar sites. At Site A, a residential rooftop in Seibersdorf, Austria, a module caught fire in March 2025: the front glass is burned through (F-01) and the back is scorched with burn spots along the cell connections (F-02). F-01 and F-02 are the front and back of the same module. The pattern points to an internal electrical fault, so other modules on that roof, from the same model and batch, are at risk. Branches from a nearby tree also hang over that array (F-04). At Site B, a ground-mount farm, tall vegetation is growing under the rows (F-03) and DC cables hang loose below the modules (F-06). At Site C, a flat roof, the modules carry a layer of dust with heavier dirt along their lower edges (F-05, F-07), which cuts output.

Key risks

- F-01, F-02: Module fire from an internal fault; the same fault could exist in other modules on the Site A roof.
- F-03: Tall, drying vegetation under the rows at Site B is a fire and shading risk.
- F-05, F-07: Soiling at Site C is reducing output every day it stays.
- F-06: Loose DC cables at Site B can be damaged by wind, animals or mowers.

Findings by category



2. Action plan

Immediate (24-72 hours)

- Site A: keep the affected string isolated; thermal-scan every other module on the roof and check connectors and the inverter's arc-fault settings before switching back on (F-01, F-02).
- Send the damaged module for failure analysis and open a warranty claim.

Next 30 days

- Site B: mow under all rows and set a regular vegetation schedule (F-03).
- Site A: trim the overhanging tree (F-04).
- Site C: clean the modules, paying attention to the lower edges (F-05, F-07).

Next 90 days

- Site B: re-secure the hanging cables with UV-rated clips (F-06).
- Re-fly all three sites, with a thermal camera at Site A, to confirm fixes.

3. Findings register

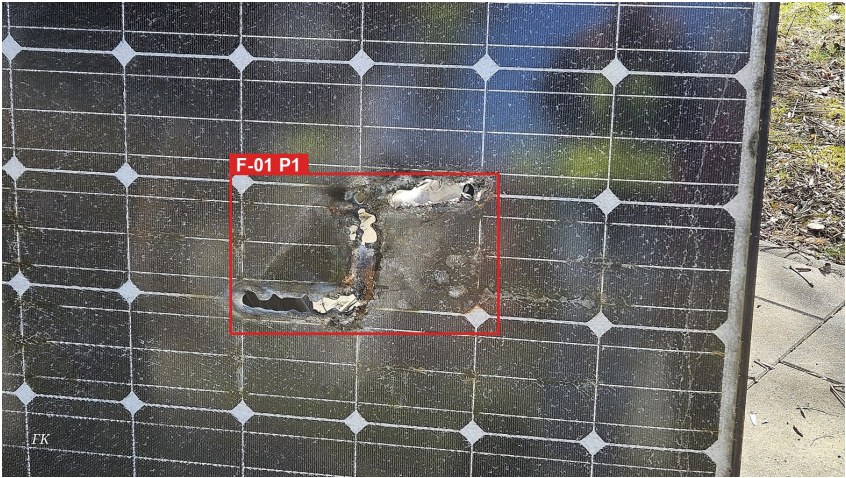
All findings, most urgent first. Each is detailed with photos and repair steps in section 4.

ID	Time	Issue	Category	Severity	Priority	Action
F-01	still	Burn-through holes in module glass	physical damage	Critical (3.9)	P1	Shut down
F-02	still	Scorched backsheet with burn marks along the cell connections	physical damage	Critical (3.9)	P1	Shut down
F-03	still	Tall vegetation under module rows	site condition	Major (2.6)	P2	Clean
F-04	still	Tree branches overhanging the array	shading	Moderate (2.2)	P2	Clean
F-05	still	Dirt build-up along module lower edges	soiling	Moderate (2.0)	P2	Clean
F-06	still	DC cables hanging loose under modules	cable issue	Moderate (1.7)	P3	Shut down
F-07	still	Dust and grime on module glass	soiling	Moderate (1.7)	P3	Clean

4. Detailed findings

F-01 Burn-through holes in module glass

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	still (seen in 2 frame(s))
Component	PV module (removed from Site A roof)
Extent	Area of about 4 cells, ~30 x 25 cm
Category	physical damage (98% confidence)
Severity	Critical, score 3.90 / 4
Safety / environmental hazard	66% probability
Likelihood it is a real defect	95%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Qualified PV technician (electrical)
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The module has burned through from behind: two holes have melted through the glass and cells, with charred and melted material around them. This is the result of an internal electrical fault (hotspot or arc) that caught fire.

Visual evidence: Two ragged holes with melted, silvered edges, brown scorching around them across 3-4 cells.

How to fix it

For this finding: Keep the module out of service and send it for failure analysis or warranty; test the rest of the string and the other modules on the roof for the same fault.

Standard procedure

- 1 Treat the module as a shock hazard: broken glass can expose live parts.
- 2 Isolate the string at the combiner box or inverter before touching it.
- 3 Replace the module with a matching model (same electrical rating); record the serial number for warranty.
- 4 Run an I-V curve or string check after replacement.

References: IEC 62446-1 (testing, documentation, maintenance); NFPA 70 / NEC Article 690

F-02 Scorched backsheet with burn marks along the cell connections

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	still (seen in 1 frame(s))
Component	PV module backsheet (removed from Site A)
Extent	Burn area ~40 x 15 cm plus spots across most of the module width
Category	physical damage (99% confidence)
Severity	Critical, score 3.88 / 4
Safety / environmental hazard	77% probability
Likelihood it is a real defect	93%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Qualified PV technician (electrical)
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The back of the module is badly scorched in a large patch, and there are many small burn spots in a line along the cell connections, some near the junction box. This points to an electrical fault inside the module, not an outside fire.
Visual evidence: Large black burned area with melted material in the middle, plus a row of small brown-black burn spots across the backsheet.

How to fix it

For this finding: Keep for failure analysis; check the same model and batch elsewhere on the roof with a thermal camera, and have an electrician inspect the inverter's arc-fault and ground-fault settings.

Standard procedure

- 1 Treat the module as a shock hazard: broken glass can expose live parts.
- 2 Isolate the string at the combiner box or inverter before touching it.
- 3 Replace the module with a matching model (same electrical rating); record the serial number for warranty.
- 4 Run an I-V curve or string check after replacement.

References: IEC 62446-1 (testing, documentation, maintenance); NFPA 70 / NEC Article 690

F-03 Tall vegetation under module rows

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	still (seen in 1 frame(s))
Component	ground under the rows, Site B
Extent	Across the visible rows
Category	site condition (93% confidence)
Severity	Major, score 2.62 / 4
Safety / environmental hazard	41% probability
Likelihood it is a real defect	93%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Civil / site maintenance
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

Grass and weeds are growing tall under and between the rows. In summer this can reach the lower module edges, shading cells, and dry grass is a fire risk next to cabling.

Visual evidence: Dense green grass and weeds covering the ground under the modules and around the posts.

How to fix it

For this finding: Mow or graze under the rows on a regular schedule, keeping vegetation well below the lowest module edge.

Standard procedure

- 1 Fix drainage or regrade eroded areas near foundations.
- 2 Cut back vegetation; repair fencing.

References: Site O&M plan

F-04 Tree branches overhanging the array

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	still (seen in 1 frame(s))
Component	tree next to Site A roof
Extent	Upper 2 rows, several modules
Category	shading (99% confidence)
Severity	Moderate, score 2.17 / 4
Safety / environmental hazard	25% probability
Likelihood it is a real defect	90%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Site maintenance
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

Branches from a nearby tree hang over the top of the array. In leaf they will shade the upper modules for part of the day.

Visual evidence: Bare branches crossing over the upper module rows.

How to fix it

For this finding: Trim the tree back so it can't shade or drop debris on the array.

Standard procedure

- 1 Trim or remove vegetation casting shade; remove debris from the array.
- 2 If a fixed structure shades the array, evaluate optimisers or re-stringing.

References: Site vegetation management plan

F-05 Dirt build-up along module lower edges

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	still (seen in 1 frame(s))
Component	rooftop PV modules, Site C
Extent	Along the lower edge of 3+ modules
Category	soiling (100% confidence)
Severity	Moderate, score 1.98 / 4
Safety / environmental hazard	5% probability
Likelihood it is a real defect	89%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	O&M cleaning crew
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

Dirt has built up in a band along the lower frame edge of the modules, where rain runs off and leaves grime behind. Edge soiling shades the bottom row of cells and cuts output more than even dust does.

Visual evidence: Grey-brown band of dirt along the bottom edge of the glass, plus speckled dust over the rest of the surface.

How to fix it

For this finding: Clean the modules, paying attention to the lower edge; check the tilt allows water to run off freely.

Standard procedure

- 1 Clean affected modules with deionised water and a soft brush; no abrasive tools or high pressure.
- 2 Clean in the early morning or evening to avoid thermal shock to hot glass.
- 3 Where bird droppings recur, install bird deterrents; review the cleaning interval.

References: Module manufacturer's cleaning guide; IEC 62446-1

F-06 DC cables hanging loose under modules

P3 · Scheduled / monitor (next maintenance cycle)



Full frame with approximate location



Zoomed detail

Timestamp	still (seen in 1 frame(s))
Component	string cabling under module rows, Site B
Extent	At least 2 rows
Category	cable issue (100% confidence)
Severity	Moderate, score 1.67 / 4
Safety / environmental hazard	26% probability
Likelihood it is a real defect	90%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Qualified PV electrician
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Cables hang in loose loops below the modules at both sides of the aisle instead of being clipped to the racking. Loose cables can be chafed by wind, chewed by animals, or caught by mowers.

Visual evidence: Dark cable loops hanging freely below the module edges on the left and right.

How to fix it

For this finding: Isolate the strings, inspect the cables for damage, and clip them to the racking with UV-rated clips.

Standard procedure

- 1 Isolate the circuit before handling.
- 2 Replace damaged cable or connectors (use the same connector brand; never mix MC4 types).
- 3 Re-secure cables with UV-rated clips off the ground and away from sharp edges; add rodent protection if chewed.

References: NFPA 70 / NEC Article 690; IEC 62446-1

F-07 Dust and grime on module glass

P3 · Scheduled / monitor (next maintenance cycle)



Full frame with approximate location

Zoomed detail

Timestamp	still (seen in 1 frame(s))
Component	rooftop PV modules, Site C
Extent	Several modules in view
Category	soiling (100% confidence)
Severity	Moderate, score 1.70 / 4
Safety / environmental hazard	5% probability
Likelihood it is a real defect	85%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	O&M cleaning crew
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

The glass on several modules has a visible layer of dust and grime spread over the surface.
Visual evidence: Matte, speckled film over the cells, clearest where the light catches it.

How to fix it

For this finding: Clean with deionised water and a soft brush; set a cleaning interval based on how quickly it builds up.

Standard procedure

- 1 Clean affected modules with deionised water and a soft brush; no abrasive tools or high pressure.
- 2 Clean in the early morning or evening to avoid thermal shock to hot glass.
- 3 Where bird droppings recur, install bird deterrents; review the cleaning interval.

References: Module manufacturer's cleaning guide; IEC 62446-1

5. Methodology and limitations

How this report was produced

- Frames were sampled from the footage (still images, 10 in total).
- Detection: each frame was examined by a vision-language model (Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)) that listed every visible defect with its location, extent and visual evidence. It did not assign severity.
- Triage: each observation was scored by TypeSafe Jev (jev-1.13.0), a System One model that returns typed answers with probabilities: defect category, severity on a 5-level rubric, first action, the probability that the defect is genuine, and the probability of an immediate safety or environmental hazard.
- Rules applied by code: observations with less than 35% probability of being genuine were dropped; P1 if expected severity is 3.5 or higher or hazard probability is 60% or higher, or the recommended action is shutdown and severity is 2.5 or higher; P2 if severity is 1.75 or higher; otherwise P3. Findings with less than 60% probability of being genuine are capped at P2 and marked 'verify on site'.
- The same defect seen in neighbouring frames was merged into one finding.
- Repair procedures come from a reviewed playbook for this asset type; the per-finding fix is model-generated.

Severity scale

- 0: Cosmetic: no functional impact, note for records only.
- 1: Minor: early-stage wear; no loss of function; fix at next scheduled maintenance.
- 2: Moderate: measurable degradation or loss of output; fix within 30-90 days.
- 3: Major: active damage or significant loss of function; fix within 7-30 days.
- 4: Critical: active leak, fire, structural failure risk or danger to people or environment; act immediately.

Limitations

- Only what is visible from the air is assessed. Internal corrosion, wall thickness, electrical faults and gas leaks without a visible sign cannot be detected from RGB footage.
- Hotspot findings need radiometric thermal footage; on RGB footage they are not reliable.
- Locations are approximate boxes on the frame, not GPS coordinates.
- AI findings can be wrong. Every P1 and P2 finding should be confirmed on site before work starts.

Image credits

- Site A photos: BFV Radkersburg, via Wikimedia Commons (Category: Damaged solar panels), CC BY-SA 2.0. Cropped and annotated.
- Site B footage: Pexels video #4324109 (Pexels License).
- Site C footage: Pexels video #7211102 (Pexels License).